F&O Derivatives Analytics Platform

One-pager report — what was built and how

Repo: github.com/varyaaggarwal/fno-derivatives-analytics-platform | **Live app:** fnoderivativesanalyticsplatform.vercel.app | **API:** fando-derivatives-analytics-platform.onrender.com/api/health

What this is. A full-stack platform that pulls option-chain data (NSE/Upstox), prices options and Greeks with Black–Scholes, solves implied volatility in reverse, renders the volatility surface, decomposes position P&L by Greek, explains the option chain in plain language, and runs a live signal engine plus historical backtester for a SuperTrend-based options-selling strategy ("DOS") on Bank Nifty weekly-expiry options.

Tech Stack

Layer	Technology	Role
Frontend	Next.js 14 (React 18) + TypeScript, Tailwind, Recharts, Plotly.js	Option chain UI, charts, 3D vol surface, DOS panel
Backend	FastAPI (Python), Uvicorn	REST API, Greeks engine, IV solver, P&L decomposer
Live data	Upstox API v2 (REST) + Market Data Feed V3 (WebSocket, protobuf), NSE scrape fallback, yfinance	Option chain, spot price, real-time ticks
Database	Supabase (PostgreSQL)	Chain snapshot cache, trade log
Math	NumPy, pandas, SciPy (Brent's method)	BSM pricing, IV solving, Greeks, SuperTrend
Deploy	Vercel (frontend), Render (backend)	Hosting
Testing	pytest	30 tests across 6 files

What's Implemented

Feature	Where
Live option chain: OI, LTP, IV per strike	/api/chain, upstox_chain.py
Greeks (Delta/Gamma/Theta/Vega) via Black-Scholes	black_scholes.py
IV solver (reverse Black-Scholes, Brent's method)	iv_solver.py
Volatility surface (3D, strike x expiry)	/api/vol-surface, VolSurface3D.tsx
Interactive P&L decomposer by Greek	pnl_decomposer.py, /pnl
Plain-language cards (PCR, max pain, IV spike)	interpretation.py
Live SuperTrend signal + strike selector (DOS)	dos_strategy.py, /api/dos/live-signal
Stop-loss monitor (initial + trailing)	/api/dos/sl-status
Historical backtest + equity curve	backtester.py, /api/dos/backtest
Trade log persisted to a database	supabase_client.py
Deployed, working full stack	Vercel + Render

A Few Things Beyond The Basics

- **Real-time WebSocket feed** – a protobuf-decoded live tick feed from Upstox as an opt-in, push-based alternative to REST polling, with graceful handling of an expired/invalid token (backs off cleanly instead of retrying forever).
- **Data provenance** – every chain response carries a structured object describing exactly which data source served it, when it was captured, and how stale it is – not just a flat mock/live flag.
- **Holiday-aware market hours** – the background poller checks NSE's actual trading-holiday calendar, not just weekday + clock time.
- **Infrastructure as code** and a small frontend keep-alive hook to avoid free-tier cold starts during a live demo.
- 30 automated tests covering pricing correctness, IV round-trips, edge cases, and failure handling.

Design Decisions & Known Limitations

- The DOS strategy's Wed/Thu weekly-expiry cadence was implemented exactly as specified, but no longer reflects live NSE mechanics as of mid-2026 – Bank Nifty weekly options were discontinued in late 2024, leaving only a monthly expiry. Flagged here rather than hidden.
- Implied volatility is solved independently (Brent's method against the observed price) rather than trusted from whatever the feed reports.
- The historical backtest runs on statistically realistic mock intraday candles, since free NSE data has no intraday history for the instrument – swappable for a paid vendor feed without touching the strategy logic itself.
- Backtest option premiums are priced at a flat assumed IV (14%), stated explicitly.
- Stated assumptions: same-day expiry for both weekly sessions, standard Bank Nifty lot size, a 6.5% risk-free rate.

Validation Performed

- Put-call parity holds to within 1e-8.
- Matches a known textbook Black-Scholes example (S=42, K=40, r=10%, sigma=20%, T=0.5y → C≈4.76).
- The IV solver round-trips: price → solved IV recovers the input volatility to 1e-4.
- Deep in/out-of-the-money delta sanity checks (→1 / →0).
- Full automated test suite: 30/30 passing.